SNU ISLAB RESEARCH · 2026

Trading the Bitcoin Cycle

Evaluating Technical, Macro, and Cycle-Based Signals in a Walk-Forward Framework

An empirical evaluation of whether Bitcoin's position within its broader market cycle contains predictive and economic information beyond conventional technical and macroeconomic indicators.

HEADLINE RESULTS

51.26%	**1.31**	**62.07%**	**−51.86%**
Out-of-sample accuracy	**Strategy Sharpe ratio**	**Strategy CAGR**	**Maximum drawdown**
Cycle model, ±3% target	vs. 0.69 buy-and-hold	vs. 26.13% buy-and-hold	vs. −81.53% buy-and-hold

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01 Executive Summary

This study tests whether Bitcoin's position within its broader market cycle contains more useful out-of-sample information than conventional technical and macroeconomic indicators. Four XGBoost specifications are evaluated within a common expanding-window walk-forward design. The final model classifies the next 30-day return as Bearish, Neutral, or Bullish using a ±3% threshold.

51.26%
Out-of-sample accuracy
Cycle model

0.401
Macro F1
Finalized ±3% target

62.07%
Strategy CAGR
Long/Cash, net of 0.10% costs

1.31
Sharpe ratio
vs. 0.69 buy-and-hold

What the Evidence Says

- Cycle-position variables outperformed the technical and macroeconomic benchmarks under an identical modeling framework and validation procedure.
- The combined model did not improve on the cycle-only model, indicating that additional predictors contributed noise rather than stable information.
- The Long/Cash strategy improved return per unit of risk primarily by reducing exposure during severe declines, not by remaining continuously invested.
- Bootstrap resampling supports the risk-adjusted advantage: the strategy's Sharpe ratio exceeded buy-and-hold in 98.6% of samples and its drawdown was better in 99.3%.
- The strongest specification is informative but not yet fully deployable, because several cycle variables depend on turning points identified in hindsight.

Bottom line

Cycle position carries genuine out-of-sample information about Bitcoin's forward return distribution, and its practical value lies in exposure management rather than point forecasting. The result is presented as evidence for cycle-aware modeling, not as a finished live-trading system.

02 Abstract

This study tests whether Bitcoin's position within its broader market cycle helps predict future 30-day returns. Four XGBoost models are compared using technical, macroeconomic, cycle-based, and combined feature sets within a walk-forward out-of-sample framework. Returns are classified as Bearish, Neutral, or Bullish using ±3% thresholds.

The cycle model performed best, reaching 51.26% out-of-sample accuracy and a Macro F1 score of 0.401. It outperformed the technical and macroeconomic models, while combining all feature groups did not improve results. The most useful predictors were measures of cycle position, including Percent Through Cycle, Days Since Halving, Days Until Top, and Days Since Bottom.

A Long/Cash strategy based on the cycle model produced a 62.07% CAGR and a Sharpe ratio of 1.31, compared with 26.13% and 0.69 for Bitcoin buy-and-hold. The strategy also reduced volatility and maximum drawdown while remaining invested 48.51% of the time.

The results suggest that cycle position contains useful information for deciding when to hold Bitcoin and when to remain in cash. However, some cycle variables rely on historical turning points that are not fully observable in real time.

Keywords

Bitcoin; market cycles; XGBoost; walk-forward validation; regime classification; systematic trading; exposure management

I. Introduction

Bitcoin is not difficult to trade because it lacks identifiable patterns. It is difficult because the value of those patterns changes across market regimes.

Since 2009, Bitcoin has developed from a niche digital currency into a globally traded risk asset. The market now includes retail participants, hedge funds, publicly listed corporations, proprietary trading firms, asset managers, and exchange-traded investment products. This institutionalization has increased liquidity and broadened market participation, but it has not made Bitcoin easier to forecast. The asset continues to exhibit severe drawdowns, abrupt changes in market structure, sharp volatility, and prolonged directional moves.

The relevant question for a trader, therefore, is not simply whether Bitcoin will generate positive returns over the long run. The more important question is whether exposure can be adjusted systematically as the distribution of forward returns changes. An investment strategy that remains fully invested across prolonged periods may capture Bitcoin's long-term appreciation, but it also remains fully exposed to drawdowns and volatility. By contrast, a model capable of distinguishing between favorable and unfavorable market conditions may improve capital efficiency and risk-adjusted returns without predicting the exact future price.

Most existing forecasting models approach this problem through either technical or macroeconomic information.

Technical models use historical price and volume data to measure momentum, volatility, trend strength, and mean reversion. Common inputs — including the Relative Strength Index, Moving Average Convergence Divergence, moving averages, Bollinger Bands, Average True Range, recent returns, and realized volatility — are widely used because they translate market behavior into systematic trading signals that can be updated continuously as new data become available.

Macroeconomic models treat Bitcoin as part of the broader global risk environment. By incorporating variables related to monetary and fiscal policy, inflation, liquidity, equity-market performance, and dollar strength, this approach has become increasingly relevant as Bitcoin grows more sensitive to institutional flows and changes in risk appetite. Variables such as the federal funds rate, money supply, Treasury yields, the U.S. Dollar Index, the S&P 500, and the Nasdaq may therefore provide information not contained in Bitcoin's own trading history.

Although both approaches are economically intuitive, they tend to focus primarily on current market conditions. Technical indicators describe what price is doing now, while macroeconomic indicators describe the environment in which capital is being allocated. The limitation is that neither necessarily captures where Bitcoin stands within the broader progression of its historical bull and bear cycles.

This distinction matters because identical signals may produce different outcomes depending on cycle position. A strong momentum reading near the beginning of a market expansion may represent the start of a durable trend. The same reading late in an extended bull market may instead reflect crowded positioning and deteriorating return asymmetry. Similarly, improving macroeconomic conditions may provide meaningful upside during an early recovery but offer limited incremental information after substantial appreciation has already occurred.

This study examines whether Bitcoin's position within its historical market cycle contains predictive information beyond conventional technical and macroeconomic signals. The analysis converts cycle structure into a set of quantitative variables, including time elapsed since an identified market bottom, progression through bull and bear phases, percentage of estimated cycle completion, time since the most recent halving, and indicators distinguishing individual cycle episodes. These variables are not intended to imply that Bitcoin follows a perfectly deterministic four-year pattern. Rather, they test whether broad cycle position changes the conditional distribution of future returns.

The empirical design compares four competing information sets. The first contains technical indicators, the second contains macroeconomic and cross-asset variables, the third contains cycle-position variables, and the fourth combines all three groups. Each specification is evaluated using the same supervised machine learning framework so that differences in performance reflect the information contained in the features rather than differences in modeling methodology.

The prediction target classifies Bitcoin's subsequent 30-day return into three market states: Bullish, Neutral, and Bearish. Model performance is evaluated using walk-forward validation, under which each prediction is generated using only information available before the test period. This structure is essential in trading research. A model that performs well using revised data, future cycle peaks, or retrospectively identified regimes may be statistically interesting but economically unusable. All cycle-related variables must therefore be constructed on an ex ante basis to prevent future information from entering the model.

Research questions

a. Does Bitcoin's position within its historical market cycle provide greater out-of-sample information about future market direction than conventional technical indicators and macroeconomic variables?

b. Can cycle-aware signals improve exposure decisions, or do they merely repackage Bitcoin's underlying long-term directional bias?

This study makes three contributions. First, it translates Bitcoin's recurring bull and bear market structure into a systematic feature set that can be tested quantitatively. Second, it compares cycle, technical, and macroeconomic information within an identical predictive framework. Third, it evaluates whether cycle-aware modeling can support more efficient market exposure by distinguishing between environments with materially different forward return distributions.

The objective is not to identify a perfect market-timing rule. No such rule is likely to remain stable across every Bitcoin regime. The objective is to determine whether cycle position provides a repeatable informational edge and whether that edge remains visible under realistic out-of-sample conditions.

II. Background and Literature Review

Research on Bitcoin forecasting has generally focused on three areas: technical signals, macro-financial variables, and machine learning methods. A smaller group of studies has examined supply halvings and longer-term market cycles. Together, this literature provides the basis for testing whether Bitcoin's position within a cycle contains information that is not already captured by more conventional predictors.

a. Technical and Macroeconomic Predictors

Technical indicators are widely used in Bitcoin forecasting because they can be calculated in real time and translated directly into trading rules. Common examples include moving averages, RSI, MACD, Bollinger Bands, ATR, lagged returns, and realized volatility. Prior research suggests that these variables can improve short-term forecasts, although their effectiveness often varies across sample periods and market conditions (Kristjanpoller & Minutolo, 2018; Liu, 2019).

One limitation is that many technical indicators are built from the same underlying price series. As a result, adding more indicators does not always introduce new information and may instead increase correlation across features. Their interpretation may also depend on the surrounding regime. A momentum signal that works during an early expansion, for example, may be much less useful late in a mature bull market.

Other studies incorporate macroeconomic and cross-asset variables such as interest rates, money supply, inflation, Treasury yields, the U.S. dollar, and major equity indices. This reflects Bitcoin's growing connection to global liquidity conditions and broader shifts in risk appetite (Bouri et al., 2017).

These variables are economically relevant, but they are not always easy to use in practice. Many are released monthly or quarterly, some are revised after publication, and all must be aligned carefully with market data. A forecasting model should only use information that would have been available at the time the prediction was made.

b. Machine Learning and Validation

Bitcoin's nonlinear return behavior has encouraged the use of methods such as Random Forests, Support Vector Machines, neural networks, and gradient boosting. These models can capture interactions and threshold effects that may be missed by linear specifications. McNally et al. (2018), for example, apply machine learning methods to Bitcoin price prediction, while Chen and Guestrin (2016) introduce the XGBoost framework used in this study.

The main risk is overfitting. Flexible models can produce strong historical results when features, parameters, and trading rules are repeatedly adjusted to the same sample, even if the underlying relationship is unstable (Bailey et al., 2014). This makes the evaluation process especially important.

For financial time series, random train-test splits are inappropriate because they can allow information from later periods to influence earlier predictions. Walk-forward validation avoids this problem by training the model only on past observations and testing it on later, unseen data. This more closely reflects the conditions under which a real trading model would operate.

c. Bitcoin Cycles and the Research Gap

Bitcoin's fixed issuance schedule provides a natural reason to study longer-term cycles. Roughly every four years, the block reward paid to miners is reduced by half, lowering the rate of new supply. These events have often occurred near major changes in Bitcoin's broader market cycle, although the small number of completed cycles makes it difficult to support any rigid four-year rule.

Most existing research treats halvings as isolated events or studies technical and macroeconomic predictors separately. Less attention has been given to Bitcoin's continuous position within a larger bull or bear cycle.

This study addresses that gap by constructing cycle-position variables and comparing them directly with technical and macroeconomic predictors using the same machine learning model and walk-forward validation procedure. The goal is to determine whether cycle position adds independent out-of-sample information or simply restates patterns already contained in conventional signals.

III. Data and Feature Construction

This section describes the dataset used in the empirical analysis, the construction of the prediction target, and the three information sets evaluated by the model. The objective is to ensure that each variable has a clear economic interpretation and that the data structure reflects the information available to a trader at the time of prediction.

a. Bitcoin Market Data

Daily Bitcoin OHLCV data (BTC-USD) are obtained from Yahoo Finance from February 2017 onward and aligned by date with macroeconomic and cross-asset series drawn from FRED. The final labeled sample contains 3,403 daily observations spanning February 19, 2017 through June 14, 2026. The cycle framework is anchored to three historical bottoms and the intervening bull and bear phases shown below.

FIGURE 1

Figure 1. Bitcoin price and the historical cycle structure used to construct the cycle-position variables. Green bands mark assumed 1,064-day bull phases; red bands mark 364-day bear phases.

b. Prediction Target

The model predicts Bitcoin's market state over the following 30 calendar days. The forward return is defined as $R_{t,t+30} = (P_{t+30} / P_t) - 1$.

Each observation is assigned to one of three mutually exclusive classes. A forward return below −3% is labeled Bearish, a return between −3% and +3% is labeled Neutral, and a return above +3% is labeled Bullish. The neutral band prevents small price changes from being treated as meaningful directional regimes.

FIGURE 2

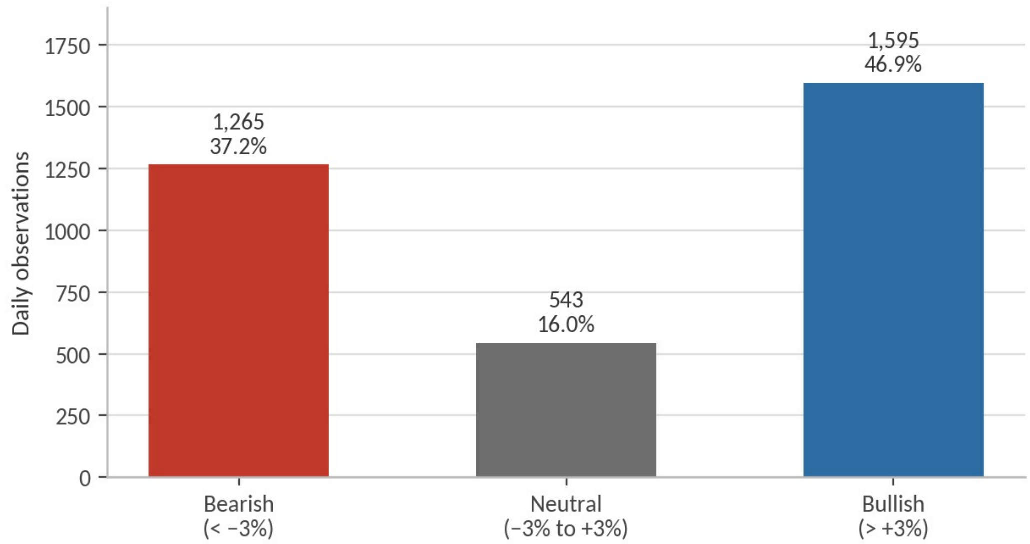

Figure 2. Distribution of the 30-day Bullish, Neutral, and Bearish target classes across the labeled sample.

The Bullish class represents 46.9% of the final sample, the Bearish class 37.2%, and the Neutral class 16.0%. This imbalance is relevant when interpreting classification accuracy and motivates the use of macro-averaged F1 scores alongside overall accuracy.

c. Technical Features

The technical information set is constructed entirely from Bitcoin's historical price and volume data. The variables are intended to capture trend, momentum, volatility, trading activity, and short-term price action. The final technical specification contains 23 features.

TABLE 1 **Technical feature groups**

Category	Variables
Trend	EMA20, EMA50, MACD, MACD signal, MACD histogram
Momentum	RSI and stochastic oscillator
Volatility	Bollinger Bands, Bollinger width, ATR, 7-day and 30-day realized volatility
Volume	On-Balance Volume and daily volume change
Price action	Daily return, range, percentage range, candle body, upper wick, and lower wick

Table 1. Technical predictors grouped by economic role. Full definitions appear in Appendix A.1.

These indicators are calculated using only current and lagged market observations, so no future information enters the feature set.

d. Macroeconomic and Cross-Asset Features

The macroeconomic information set contains the Federal Funds Rate, Consumer Price Index, M2 money supply, the 10-year U.S. Treasury yield, the S&P 500, the Nasdaq Composite, and the U.S. Dollar Index. The variables represent monetary policy, inflation, liquidity, interest rates, equity-market risk appetite, and dollar strength.

Because the series are observed at different frequencies, each macroeconomic variable is aligned with the daily Bitcoin dataset and forward-filled until a new observation becomes available. Forward-filling allows each trading date to be matched with the most recently recorded value but does not create additional independent macroeconomic information.

The analysis uses currently available historical series rather than archived real-time data vintages. Macroeconomic revision bias therefore cannot be eliminated completely and is treated as a limitation of the empirical design.

e. Cycle Features

The cycle information set translates Bitcoin's broad historical bull and bear structure into numerical variables. Three historical cycle bottoms are used: January 14, 2015; December 15, 2018; and November 21, 2022. The framework assumes an approximate bull-phase duration of 1,064 days and a bear-phase duration of 364 days, producing an estimated full-cycle length of 1,428 days.

TABLE 2 **Cycle-position feature definitions**

Feature	Definition
Days Since Bottom	Calendar days elapsed since the assigned historical cycle bottom
Days Until Top	Days remaining until the assumed 1,064-day bull-phase endpoint
Bull Phase	Indicator identifying observations within the assumed bull phase
Percent Through Cycle	Share of the estimated 1,428-day cycle completed
Days Since Halving	Calendar days elapsed since the most recent Bitcoin halving
Cycle Number	Identifier distinguishing the historical cycle episodes
Bull Progress	Bull-phase completion, bounded between zero and one
Bear Progress	Bear-phase completion, bounded between zero and one

Table 2. Cycle predictors. Halving dates used are July 9, 2016; May 11, 2020; and April 20, 2024.

These variables do not assume that Bitcoin follows a perfectly deterministic four-year schedule. Instead, they test whether broad cycle position changes the conditional distribution of future returns.

> **Identification issue**
>
> Several cycle variables rely on historical market bottoms that can be identified with certainty only after the fact. The full cycle specification should therefore be interpreted as a test of whether retrospective cycle position

> contains information, not as a completely deployable real-time signal. A stricter robustness specification uses only Days Since Halving and Cycle Number, which are based on objectively observable dates; it is reported in Section VI.h.

f. Preprocessing and Leakage Controls

All observations are sorted chronologically before estimation. Infinite values are replaced with missing values, and rows containing unavailable predictors or missing 30-day forward returns are removed. The forward return is used only to construct the dependent variable and is never included as a predictor.

Model evaluation follows a walk-forward structure in which each test period is predicted using only observations from earlier periods. Trading signals are shifted forward by one day before strategy returns are calculated, preventing the model from earning a return using information that would not yet have been available at execution.

No feature scaling is required for XGBoost because tree-based algorithms form decision rules through variable thresholds rather than distance-based comparisons.

IV. Methodology

This section explains how the competing information sets are evaluated, how the model is trained, and how statistical predictions are translated into trading positions.

a. Model Design

The empirical analysis uses Extreme Gradient Boosting (XGBoost) to classify Bitcoin's subsequent 30-day return as Bullish, Neutral, or Bearish. XGBoost is appropriate for this setting because it can capture nonlinear relationships and interactions without requiring a predetermined functional form.

Four specifications are estimated: a technical model, a macroeconomic and cross-asset model, a cycle-position model, and a combined model containing all predictor groups. The same algorithm and validation procedure are applied to each specification so that differences in performance can be attributed more directly to the information contained in each feature set.

FIGURE 3

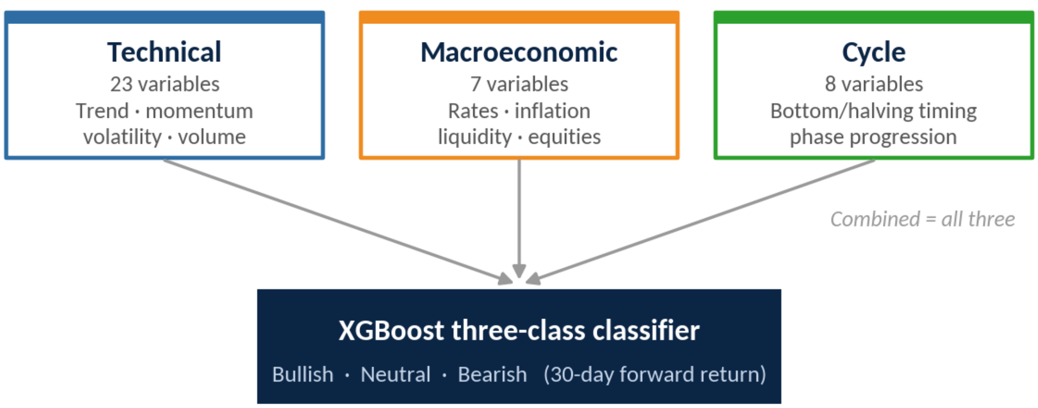

Figure 3. Predictor groups evaluated within the common XGBoost classification framework. Only the input set changes across specifications.

b. XGBoost Specification

The model uses a multiclass probability objective and assigns each observation to the class with the highest predicted probability. A fixed random seed is used to make the results reproducible.

TABLE 3 Baseline XGBoost parameters

Parameter	Value
Number of trees	300
Maximum depth	4

Parameter	Value
Learning rate	0.05
Observation subsample	0.80
Feature subsample	0.80
Random seed	42

Table 3. Identical parameters are applied across all four feature sets to preserve comparability.

Alternative parameter choices are treated as sensitivity tests rather than as part of the primary model comparison.

c. Walk-Forward Validation

Model performance is evaluated through expanding-window walk-forward validation. For each test year, the model is trained using all observations from earlier years and then applied to the following unseen year. The training window expands as additional data become available.

This procedure preserves chronological order and prevents observations from future market regimes from entering the training sample. It also approximates how the model would be updated in practice.

FIGURE 4

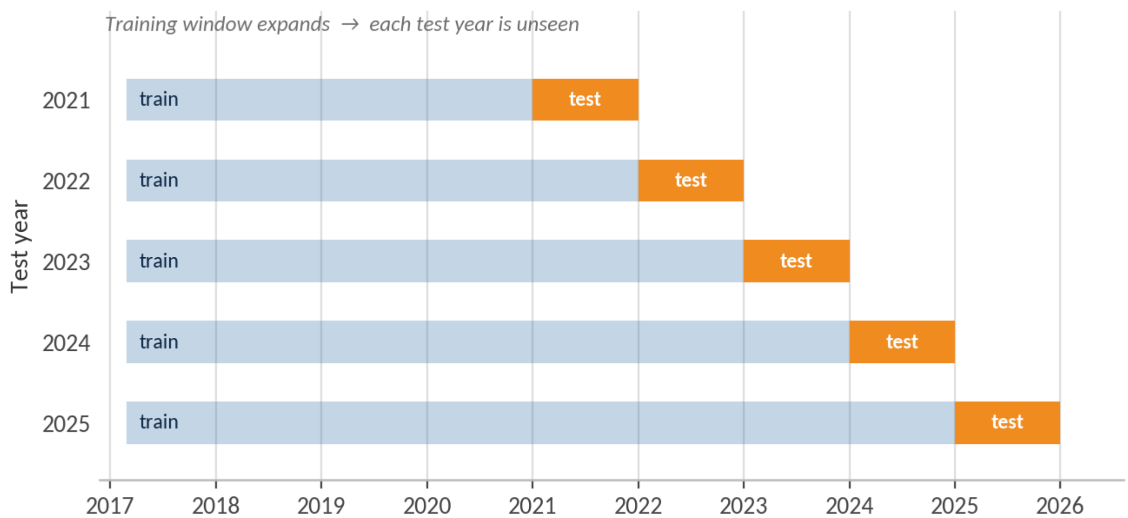

Figure 4. Expanding-window walk-forward validation. Each test year is predicted using only observations from earlier periods.

d. Evaluation Metrics

Predictive performance is evaluated using classification accuracy and macro-averaged F1. Accuracy measures the proportion of observations assigned to the correct market state: Accuracy = Correct predictions / Total predictions.

Because the Neutral class is smaller than the Bullish and Bearish classes, accuracy alone can overstate performance. Macro F1 calculates the F1 score separately for each class and gives all three classes equal weight: Macro F1 = $(F1_0 + F1_1 + F1_2) / 3$. Confusion matrices and class-level precision and recall are also examined.

e. Target-Threshold Selection

The initial specification used a ±5% neutral band for the subsequent 30-day return. A narrower ±3% threshold was later evaluated using the same walk-forward procedure. The ±3% specification was retained as the final model because it produced a less restrictive directional classification and stronger out-of-sample results. The original ±5% specification is reported as a sensitivity test to make the threshold-selection process transparent.

f. Trading-Signal Construction

To assess economic value, model predictions are converted into a long-or-cash strategy. The baseline rule takes a full long position following a Bullish prediction and moves to cash following a Bearish prediction. The treatment of Neutral predictions is specified explicitly for each strategy test.

The prediction generated on date t is shifted forward by one day before it is applied to returns. This prevents the strategy from using a signal and earning the return from the same observation. Transaction costs of 0.10% are applied when the position changes:

$$\text{Strategy return}(t+1) = \text{Position}(t) \times \text{BTC return}(t+1) - \text{Cost} \times |\text{Position}(t) - \text{Position}(t-1)|$$

Economic performance is evaluated using CAGR, annualized volatility, Sharpe ratio, maximum drawdown, market exposure, and turnover. Bitcoin buy-and-hold is used as the primary benchmark.

V. Results

This section reports the complete sequence of empirical findings. It first presents the original four-model benchmark, which compares technical, macroeconomic, cycle, and combined information sets using a ±5% target. It then presents the finalized ±3% cycle specification and its direct macroeconomic benchmark. The two stages are labeled separately to prevent results from different target definitions from being treated as a single experiment.

a. Four-Model Benchmark

The initial benchmark compares all four information sets over the 2021–2025 walk-forward period. Under the original ±5% target, the cycle model achieved the strongest average yearly accuracy and Macro F1. The macro model ranked second, while technical indicators performed worst. Combining all variables did not improve performance.

FIGURE 5

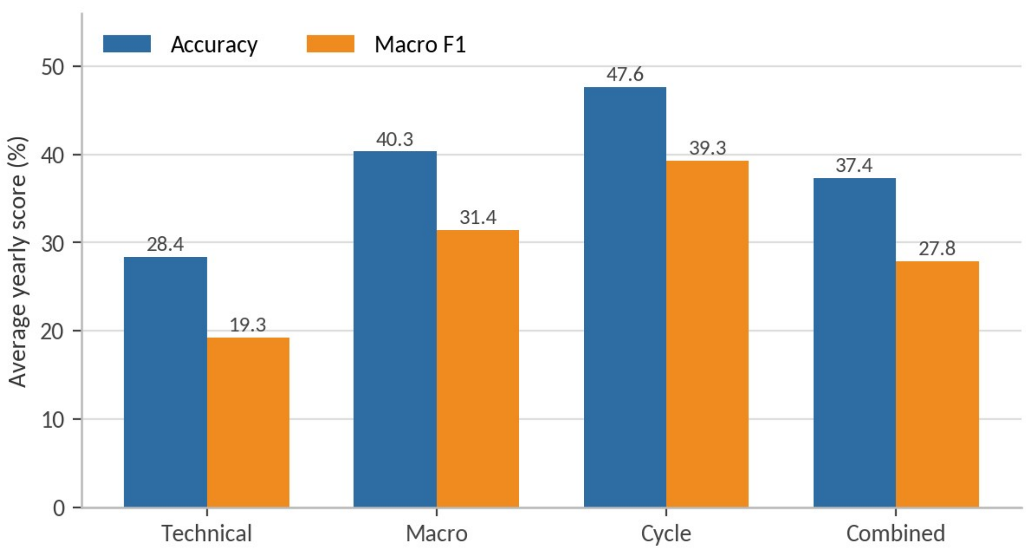

Figure 5. Average yearly walk-forward performance of all four models under the original ±5% target.

b. Walk-Forward Performance by Year

FIGURE 6

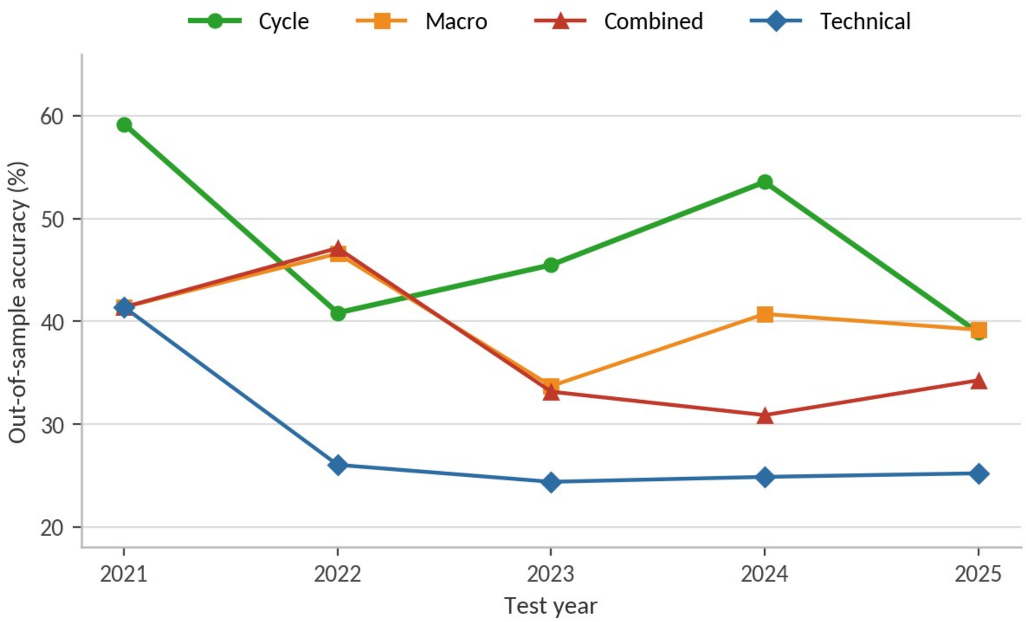

Figure 6. Annual walk-forward accuracy for the technical, macroeconomic, cycle, and combined models.

TABLE 4 **Walk-forward accuracy and Macro F1 by test year (±5% target)**

Year	Technical	Macro	Cycle	Combined
2021	0.4137 / 0.195	0.4137 / 0.195	0.5918 / 0.433	0.4137 / 0.195
2022	0.2603 / 0.167	0.4658 / 0.368	0.4082 / 0.343	0.4712 / 0.309
2023	0.2438 / 0.186	0.3370 / 0.319	0.4548 / 0.368	0.3315 / 0.274
2024	0.2486 / 0.181	0.4071 / 0.313	0.5355 / 0.431	0.3087 / 0.276
2025	0.2521 / 0.234	0.3918 / 0.377	0.3890 / 0.389	0.3425 / 0.338
Mean	**0.2837 / 0.193**	**0.4031 / 0.314**	**0.4759 / 0.393**	**0.3735 / 0.278**

Table 4. Accuracy / Macro F1 by year. The cycle model leads on the five-year mean of both metrics.

The cycle model led in 2021, 2023, and 2024, while the combined model narrowly led in 2022. No model was consistently dominant in every year. The technical model remained weak after 2021, and the combined model deteriorated materially in 2023 and 2024.

c. Technical Model

The technical specification used 23 price- and volume-derived indicators. Its weak walk-forward performance suggests that these short-horizon signals did not generalize reliably across broader Bitcoin regimes.

FIGURE 7

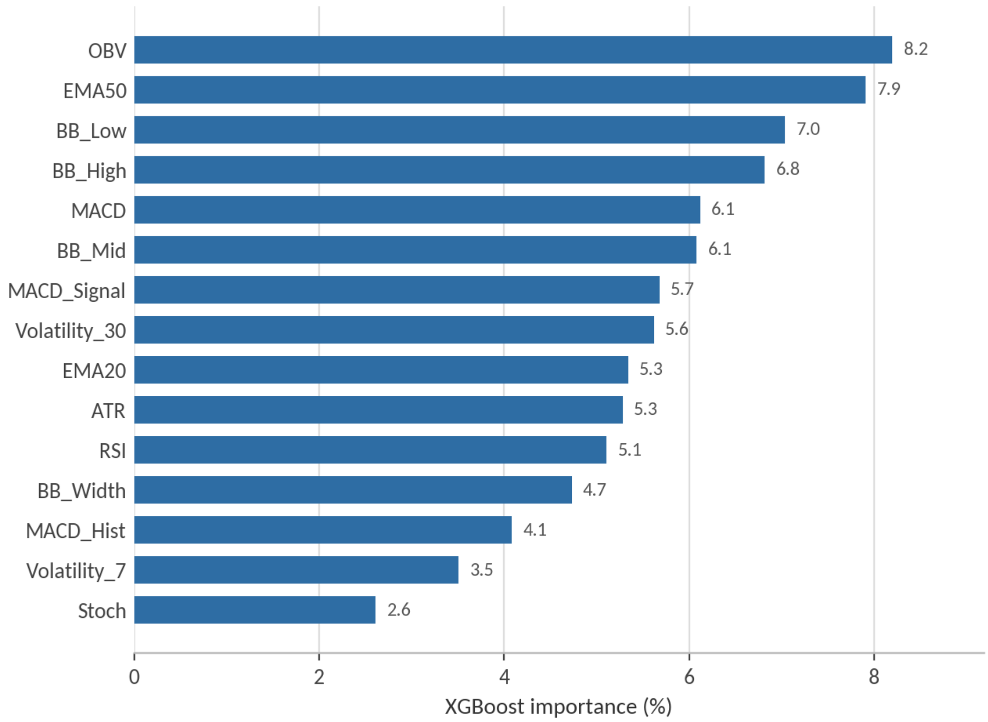

Figure 7. Fifteen highest-ranked variables in the technical model.

OBV, EMA50, Bollinger Band levels, MACD, and 30-day volatility received the greatest importance. Importance was broadly distributed rather than concentrated in one indicator. Because many technical features are transformations of the same price series, the rankings measure model usage rather than independent causal effects.

d. Macroeconomic Model

The macroeconomic specification outperformed the technical model. M2 and the Federal Funds Rate were the two leading predictors, followed by CPI, the S&P 500, the 10-year Treasury yield, Nasdaq, and DXY.

FIGURE 8

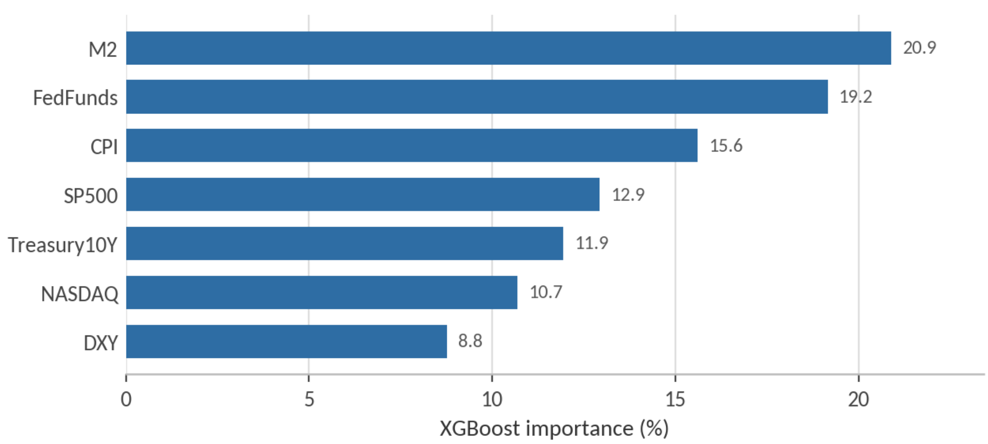

Figure 8. Feature importance within the macroeconomic model.

The result is consistent with Bitcoin behaving as a liquidity-sensitive risk asset. However, macroeconomic data are lower-frequency and may adjust too slowly to abrupt changes in market state.

e. Cycle Model: Original Benchmark

FIGURE 9

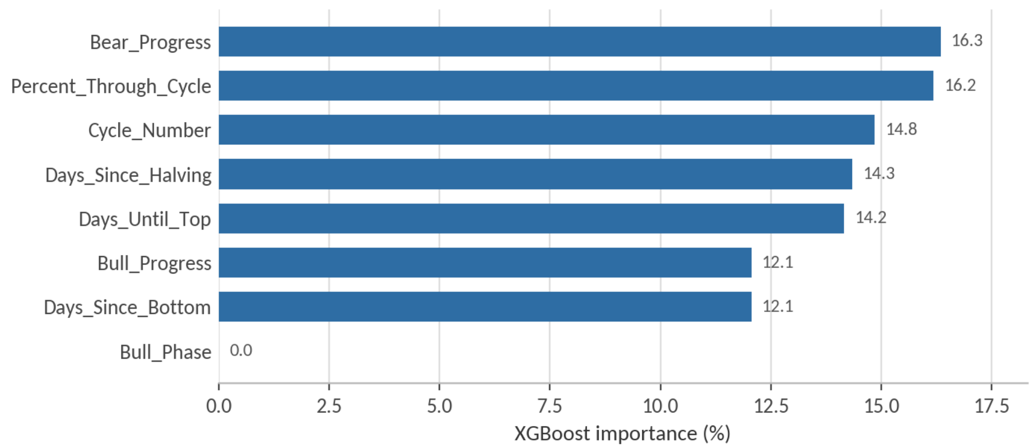

Figure 9. Cycle-feature importance in the original four-model benchmark.

Bear Progress, Percent Through Cycle, Cycle Number, Days Since Halving, and Days Until Top all received substantial weight. The binary Bull Phase variable received essentially no importance, indicating that continuous progression variables were more informative than a simple phase label.

f. Combined Model

The combined model included technical, macroeconomic, and cycle variables. Cycle variables remained prominent after all predictors were introduced, but the model still underperformed the cycle-only specification.

FIGURE 10

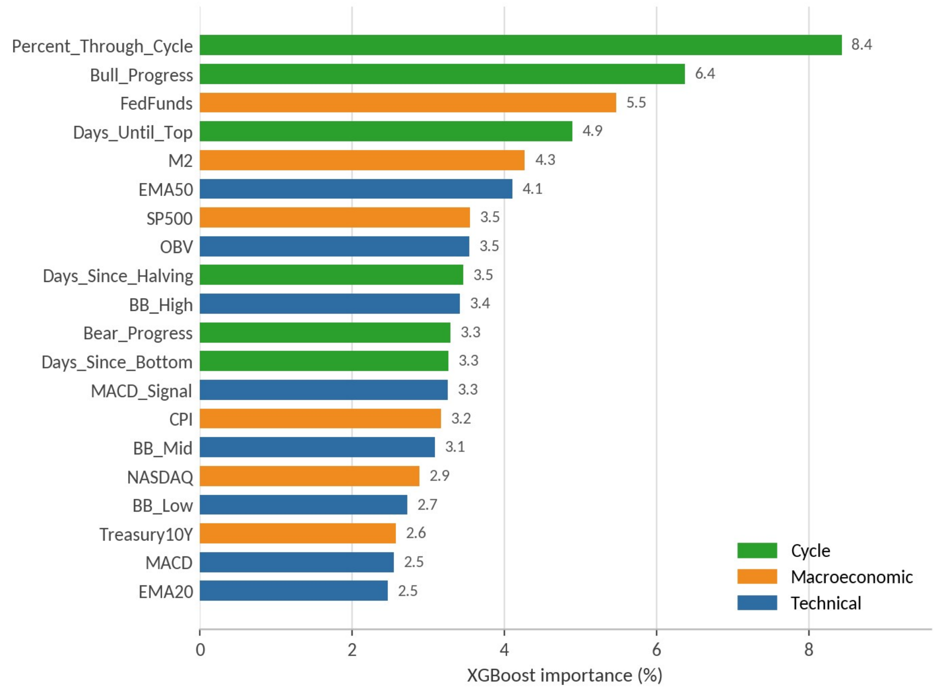

Figure 10. Twenty highest-ranked variables in the combined model, coloured by information set.

Percent Through Cycle was the most important combined-model variable, followed by Bull Progress, the Federal Funds Rate, Days Until Top, M2, and EMA50. The failure of the combined model shows that a variable can receive importance inside a fitted model without improving out-of-sample performance.

g. Finalized ±3% Cycle Model

The target was later narrowed from ±5% to ±3%. Under the finalized specification, the pooled 2021–2025 cycle model achieved 51.26% accuracy, a Macro F1 score of 0.401, and log loss of 1.500. The macroeconomic benchmark achieved 41.62% accuracy, 0.359 Macro F1, and a log loss of 1.762.

FIGURE 11

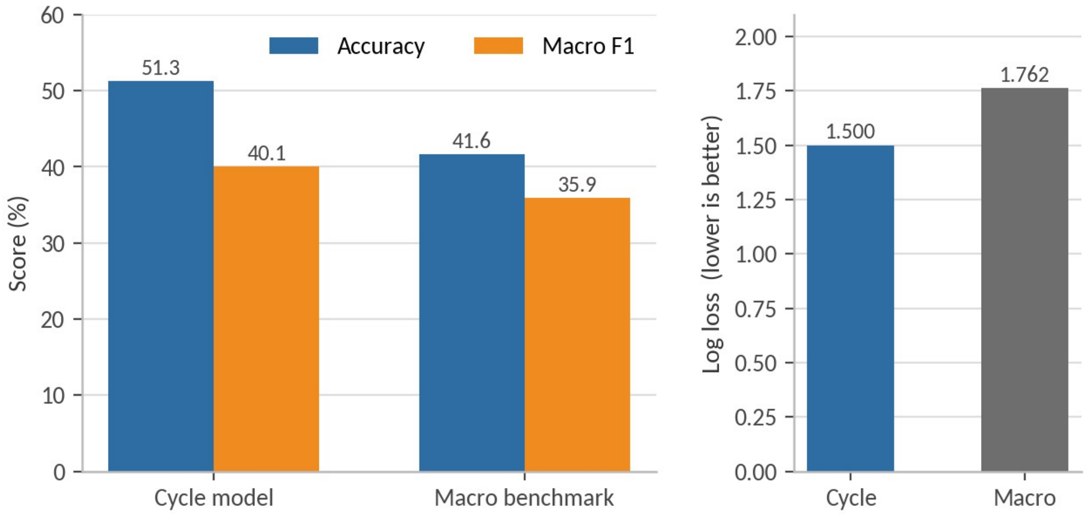

Figure 11. Finalized ±3% cycle and macroeconomic model comparison. Lower log loss indicates better-calibrated probabilities.

Statistical significance

The Diebold–Mariano statistic was 5.875 with $p < 0.000001$, and McNemar's statistic was 43.75 with $p = 3.73 \times 10^{-11}$. Both tests reject equal predictive performance in favour of the cycle model.

h. Final Cycle Class-Level Performance

FIGURE 12

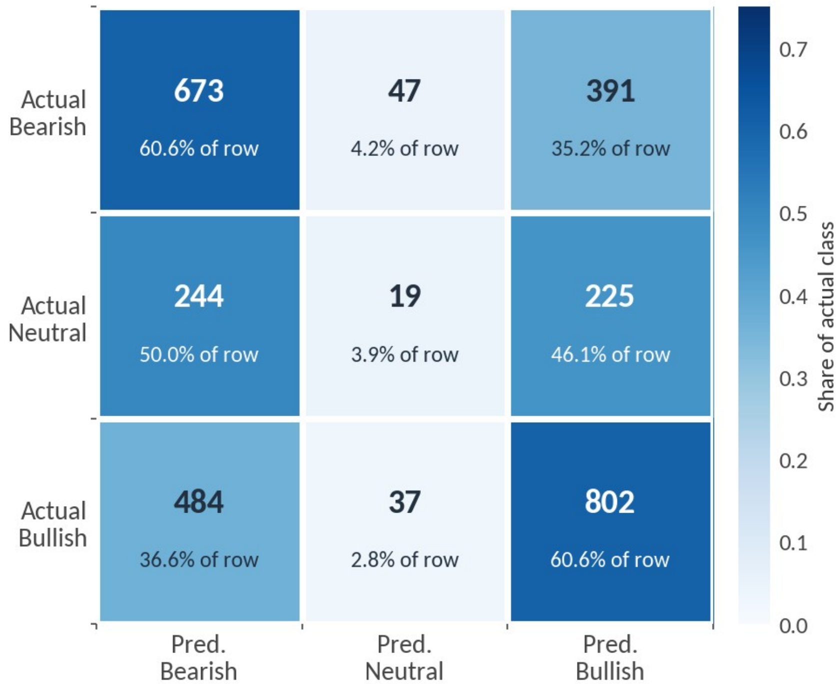

Figure 12. Confusion matrix for the finalized ±3% cycle model. Cell shading shows the share of each actual class.

The model performed substantially better on directional market states than on Neutral conditions. Bullish was the strongest class, with an F1 score of 0.585, followed by Bearish at 0.536. Neutral performance was weak, with recall of only 3.9%, indicating that the model generally assigned low-magnitude return periods to one of the directional classes. Consequently, the macro-average F1 score of 0.395 was materially lower than overall accuracy.

i. Final Cycle Feature Importance

FIGURE 13

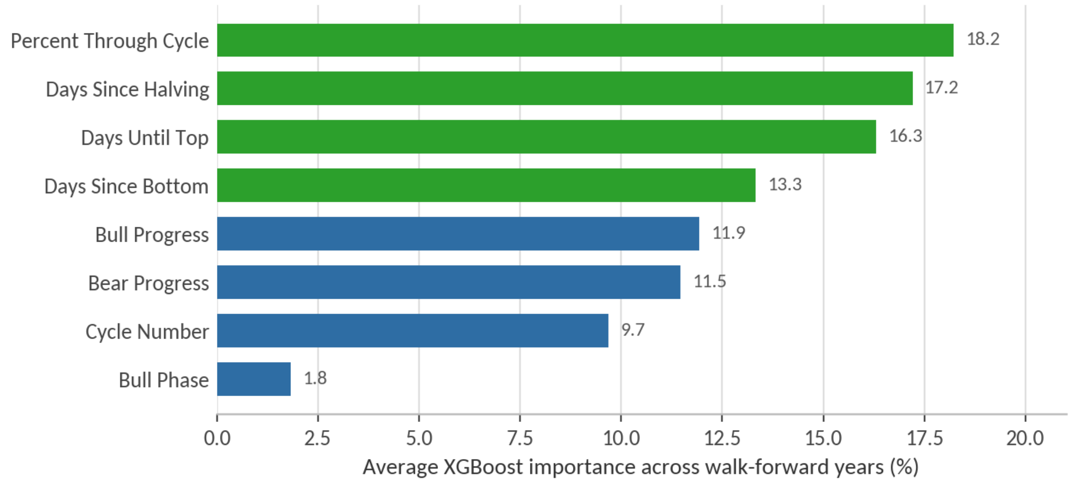

Figure 13. Average feature importance in the finalized cycle model across walk-forward years.

Percent Through Cycle ranked first, followed by Days Since Halving, Days Until Top, and Days Since Bottom. This ordering supports the interpretation that the model relies on broad temporal position rather than one isolated countdown variable.

j. Ablation Analysis

A leave-one-feature-out test evaluated whether the final cycle result depended on one variable. Removing Days Since Halving generated the largest decline in accuracy. Several other removals slightly improved accuracy while reducing Sharpe or CAGR, showing that predictive and economic value are not identical. Full results appear in Appendix D.2.

k. Forecast Confidence and Forward Returns

Predicted Bullish probability had a correlation of 0.275 with the subsequent 30-day return. The OLS coefficient was 0.119 with $p < 0.001$ and R-squared of 0.076. A ten-percentage-point increase in predicted Bullish probability was therefore associated with roughly a 1.2-percentage-point increase in the subsequent 30-day return.

FIGURE 14

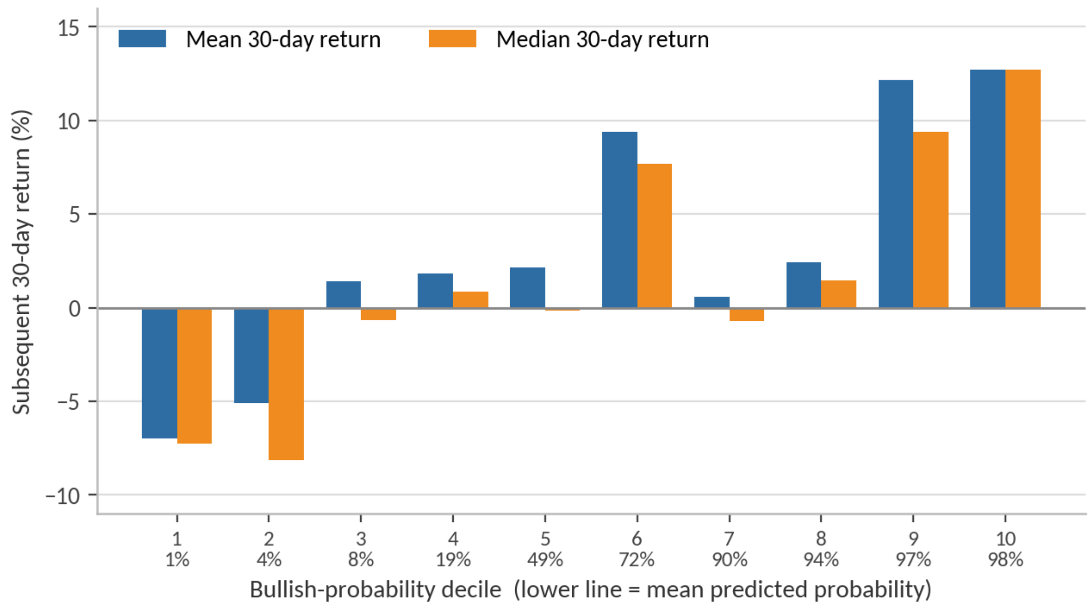

Figure 14. Mean and median subsequent 30-day returns across Bullish-probability deciles.

The relationship was not perfectly monotonic, but the highest two probability deciles generated average 30-day returns above 12% and Bullish frequencies near 67%. Observation counts were similar across deciles, although overlapping 30-day horizons create serial dependence.

I. Summary of Findings

The results provide consistent evidence that Bitcoin cycle-position variables contain more out-of-sample predictive information than conventional technical indicators and, in the finalized specification, also outperform the macroeconomic benchmark. In the original four-model comparison, the cycle model produced the strongest average walk-forward accuracy and Macro F1, while the technical model performed worst. The combined model failed to improve on the cycle-only specification, suggesting that the addition of technical and macroeconomic variables introduced noise or unstable relationships rather than incremental predictive value. This finding is reinforced by the combined-model feature-importance results, in which cycle variables remained among the highest-ranked predictors despite the model's weaker overall performance.

Under the finalized ±3% target, the cycle model achieved 51.26% pooled out-of-sample accuracy, a Macro F1 score of 0.401, and lower log loss than the macroeconomic benchmark. Statistical tests also rejected equal predictive performance in favour of the cycle model. At the class level, the model was substantially more effective at identifying Bullish and Bearish regimes than Neutral conditions. Bullish produced the strongest F1 score, followed by Bearish, while Neutral recall was only 3.9%. This indicates that the model is primarily effective as a directional regime classifier and remains limited in distinguishing low-magnitude return periods from nearby Bullish or Bearish states.

The feature-importance and ablation results suggest that predictive value is distributed across several measures of cycle position rather than concentrated in a single variable. Percent Through Cycle, Days Since Halving, Days Until Top, and Days Since Bottom consistently ranked among the most important features. In addition, predicted Bullish probability was positively related to subsequent 30-day returns, with the highest-confidence probability deciles generating the strongest average realized returns. Taken together, the findings suggest that cycle-position variables are useful not only for classifying future market direction but also for ranking the strength of potential Bullish opportunities. Section VI evaluates whether this predictive information can be converted into economically meaningful portfolio outperformance after transaction costs and risk are considered.

VI. Trading Strategy and Backtest

This section evaluates whether the cycle model's predictive signal can be converted into economically meaningful out-of-sample performance. The baseline implementation uses a one-day execution lag and a transaction cost of 0.10% per unit of turnover. All reported results are calculated from the finalized ±3% cycle-model backtest.

a. Strategy Construction

Two model-driven strategies are examined. The Long/Cash strategy holds Bitcoin only when the model predicts the Bullish class and otherwise remains in cash. The Long/Short strategy holds Bitcoin when the model predicts Bullish, holds cash when it predicts Neutral, and takes a short position when it predicts Bearish. Signals generated on day t are implemented on day t+1, preventing same-day look-ahead.

Daily strategy returns equal the lagged position multiplied by the daily Bitcoin return, less 0.1% times the absolute change in position. Buy-and-hold remains continuously invested and serves as the passive benchmark.

b. Baseline Performance

TABLE 5 **Baseline performance, January 2018 – June 2026**

Metric	Buy & Hold	Long/Cash	Long/Short
Total return	540.75%	4,665.33%	7,430.91%
CAGR	26.13%	62.07%	71.60%
Annualized volatility	64.74%	44.62%	63.70%
Sharpe ratio	**0.69**	**1.31**	**1.17**
Sortino ratio	0.92	1.23	1.56
Maximum drawdown	**−81.53%**	**−51.86%**	**−79.32%**
Calmar ratio	0.32	1.20	0.90
Win rate	51.18%	53.07%	51.95%
Market exposure	100.00%	48.51%	96.47%
Turnover units	0	39	84

Table 5. Out-of-sample backtest results net of 0.10% per unit of turnover, with a one-day execution lag.

The 3% Cycle Long/Cash strategy generated a total return of 4,665.33% and a CAGR of 62.07%, compared with 540.75% and 26.13% for buy-and-hold. Its Sharpe ratio was 1.31 versus 0.69 for the benchmark, while annualized volatility fell from 64.74% to 44.62%. Maximum drawdown improved from −81.53% to −51.86%.

The Long/Short strategy produced the highest total return and CAGR, but its Sharpe ratio was lower than Long/Cash and its maximum drawdown remained close to buy-and-hold. This indicates that short exposure increased return potential but also reintroduced substantial downside risk. The Long/Cash strategy therefore provides the cleaner risk-adjusted result and is used as the primary specification throughout.

c. Cumulative Equity Performance

FIGURE 15

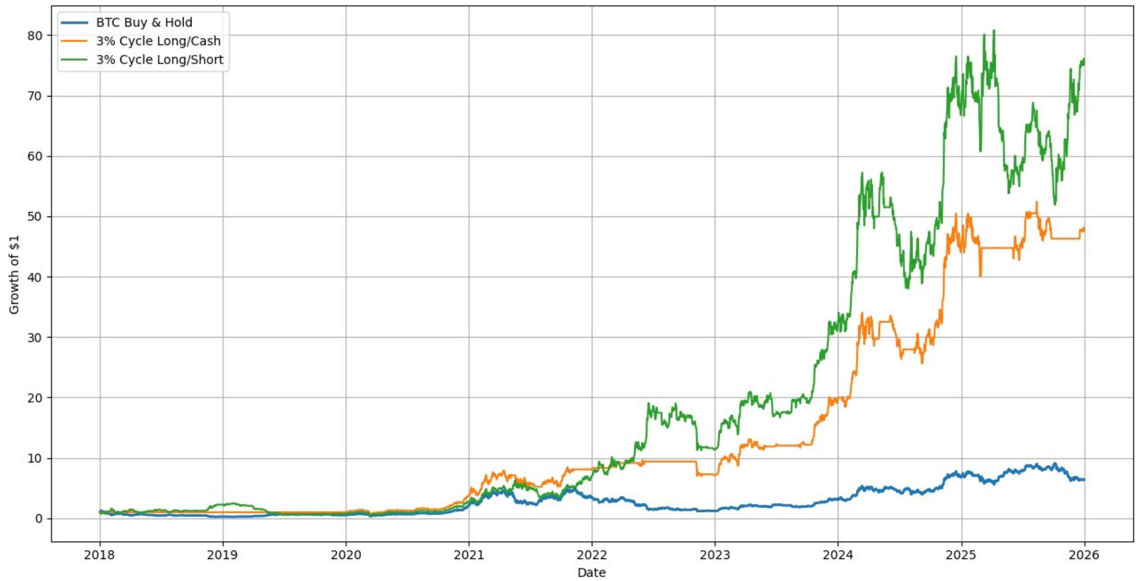

Figure 15. Growth of $1 for buy-and-hold, the 3% Cycle Long/Cash strategy, and the 3% Cycle Long/Short strategy.

The cumulative equity curve shows that the model's advantage was not generated by a single isolated trade. Long/Cash began separating materially from buy-and-hold during the 2020–2021 advance, preserved more capital during the 2022 decline, and compounded strongly during the 2023–2025 period.

d. Drawdown Behaviour

FIGURE 16

Figure 16. Drawdown paths for buy-and-hold, Long/Cash, and Long/Short.

The central benefit of the Long/Cash rule is drawdown control. By moving to cash during non-Bullish predictions, the strategy avoided a substantial portion of Bitcoin's deepest losses. Long/Short did not provide the same protection because incorrect Bearish predictions can generate losses during rapid upward reversals.

e. Year-by-Year Performance

FIGURE 17

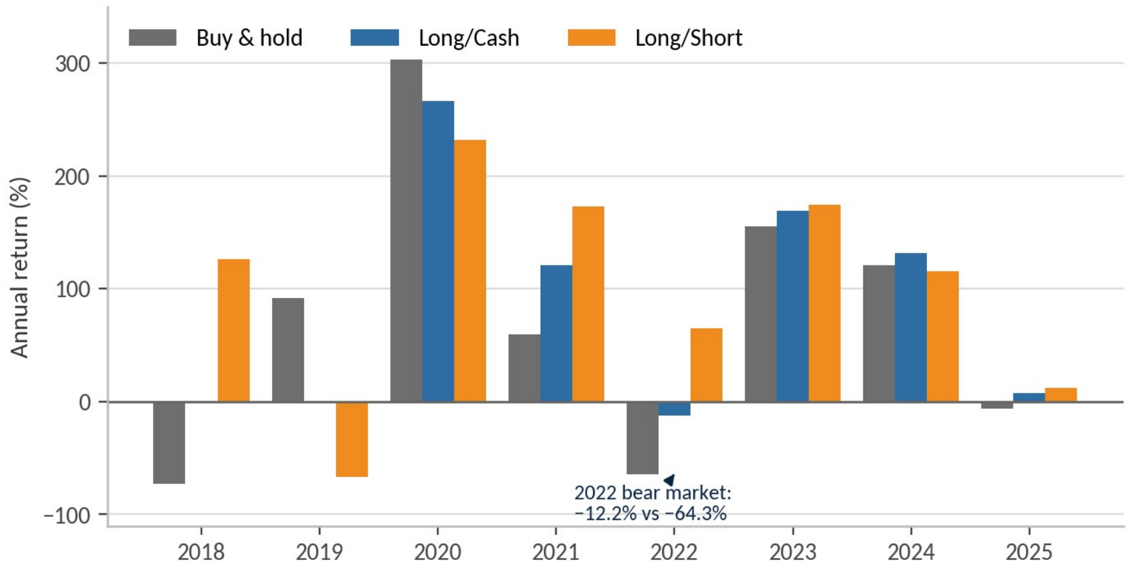

Figure 17. Calendar-year out-of-sample returns for buy-and-hold and the two model-driven strategies.

TABLE 6 **Calendar-year returns, Sharpe, and exposure**

Year	Buy & Hold	Long/Cash	Long/Short	L/C Sharpe	L/C Exposure
2018	−72.60%	0.00%	126.57%	—	0.00%
2019	92.20%	0.00%	−66.50%	—	0.00%
2020	303.16%	266.43%	231.70%	2.19	98.36%
2021	59.67%	121.00%	173.28%	1.45	76.71%
2022	**−64.27%**	**−12.15%**	**65.19%**	**−0.21**	**18.08%**
2023	155.42%	169.14%	174.22%	2.75	70.96%
2024	121.05%	131.96%	115.97%	2.08	76.78%
2025	−6.34%	7.30%	11.87%	0.40	46.85%

Table 6. The Long/Cash strategy was inactive in 2018–2019 because the walk-forward model had not yet produced an active Bullish position.

The strategy captured most of the 2020 advance, outperformed buy-and-hold in 2021, limited the 2022 loss to −12.15% versus −64.27%, and exceeded the benchmark again in 2023, 2024, and 2025. The 2022 result is especially important because avoiding a major bear market materially improves long-run compounding.

f. Signal Persistence, Exposure, and Turnover

The finalized model changed predicted class 56 times, implying an approximate average signal duration of 52.16 days. Predicted states were 47.93% Bearish, 3.53% Neutral, and 48.55% Bullish. After applying the one-day lag, Long/Cash market exposure was 48.51%, with 39 turnover units over the full sample.

56	52.2	48.51%	39
Class changes	**Days per signal**	**Market exposure**	**Turnover units**
over eight years	average duration	time invested	full sample

The strategy therefore achieved a 62.07% CAGR while being invested less than half the time. This lower exposure explains part of the volatility reduction, but not the entire result: the strategy also improved return per unit of risk, as reflected in its Sharpe and Calmar ratios.

g. Threshold Sensitivity

TABLE 7 **Sensitivity of predictive and economic performance to the neutral-band threshold**

Threshold	OOS accuracy	CAGR	Sharpe ratio
0%	55.73%	26.09%	0.765
3%	**51.11%**	**62.07%**	**1.310**
5%	46.08%	56.31%	1.249
10%	44.57%	30.84%	0.875

Table 7. The 0% threshold maximises classification accuracy; the 3% threshold maximises economic performance.

FIGURE 18

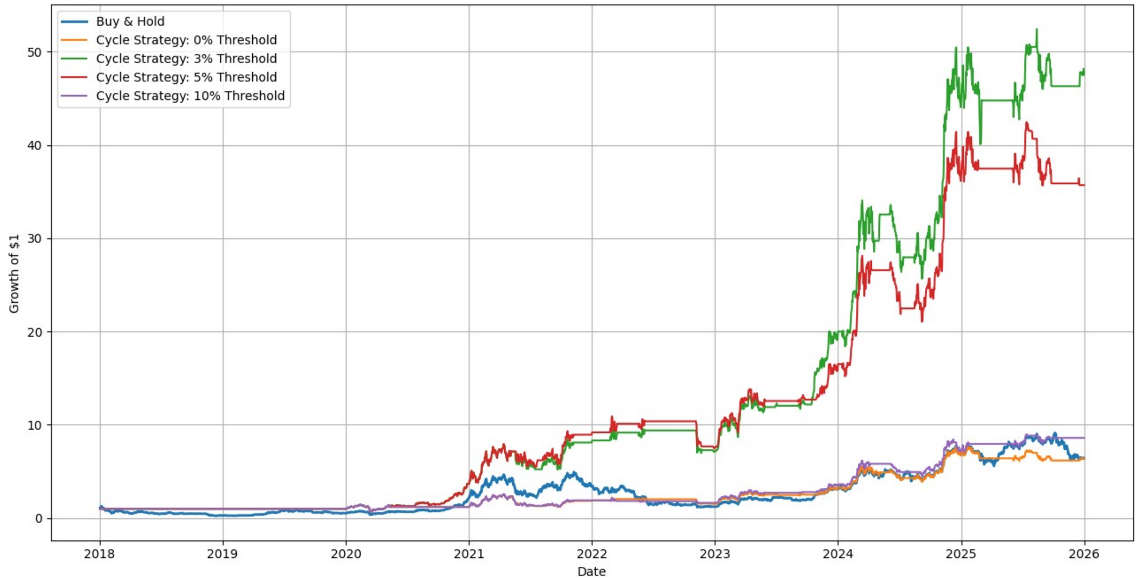

Figure 18. Cumulative strategy performance across 0%, 3%, 5%, and 10% regime thresholds.

FIGURE 19

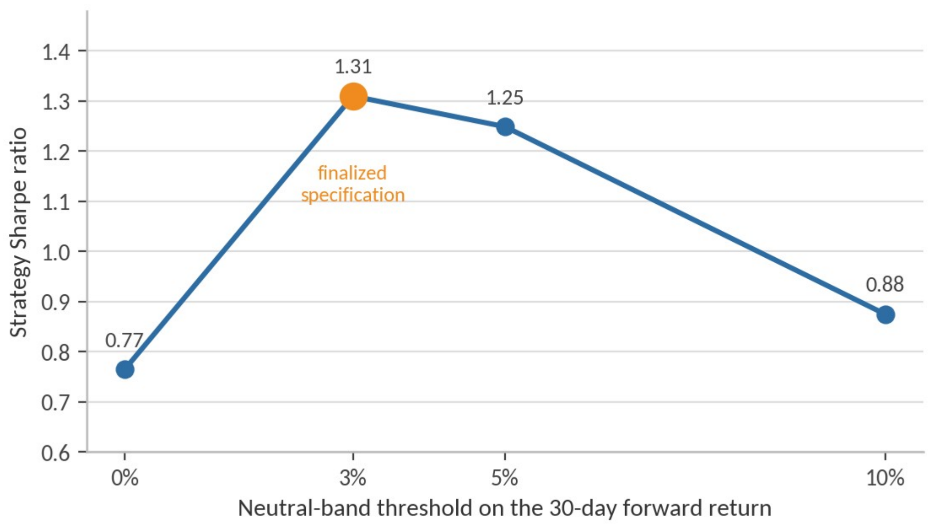

Figure 19. Strategy Sharpe ratio by target threshold.

The 3% threshold produced the strongest overall economic performance, with a 62.07% CAGR and a Sharpe ratio of 1.31. The 5% threshold remained competitive, while the 10% threshold reduced both CAGR and Sharpe. The 0% specification achieved the highest classification accuracy but much weaker trading performance. This demonstrates that higher predictive accuracy does not automatically imply greater investment value when many correctly classified observations represent economically small returns.

h. Halving-Only Robustness

The halving-only specification uses only Days Since Halving and Cycle Number, avoiding retrospective market-bottom information.

TABLE 8 Halving-only specification versus the full cycle model

Strategy	Total return	CAGR	Sharpe	Max drawdown	Exposure
Buy & Hold	540.75%	26.13%	0.69	−81.53%	100.00%
Full cycle (3%)	**4,665.33%**	**62.07%**	**1.31**	**−51.86%**	**48.51%**
Halving-only (3%)	637.32%	28.36%	0.82	−51.86%	39.03%

Table 8. The halving-only variant uses only objectively observable dates and no retrospective turning points.

It achieved a 28.36% CAGR and a Sharpe ratio of 0.82, modestly exceeding buy-and-hold on a risk-adjusted basis but materially underperforming the full cycle model. This result suggests that objectively observable halving timing contains useful information, while the stronger full-cycle result partly depends on richer bottom-based cycle-position measures.

i. Bootstrap Significance

Moving-block resampling preserves short-run time dependence and provides a more realistic test of whether the Long/Cash strategy's performance advantage could arise by chance.

TABLE 9 Moving-block bootstrap summary (n = 5,000)

Metric	Buy & Hold	Strategy	Observed difference	95% CI for difference
CAGR	26.13%	62.07%	+35.94 pp	−15.48 pp to +77.70 pp
Sharpe ratio	0.68	1.31	+0.62	+0.067 to +1.23
Maximum drawdown	−81.53%	−51.86%	+29.67 pp	+4.65 pp to +55.45 pp

Table 9. Percentage-point differences are shown for CAGR and maximum drawdown.

Comparison	Share of samples	One-sided p-value	Interpretation
Strategy CAGR > buy-and-hold	92.18%	0.0784	Positive, not significant at 5%
Strategy Sharpe > buy-and-hold	98.64%	0.0138	Statistically significant
Strategy drawdown better	99.28%	0.0074	Statistically significant

Table 10. Bootstrap outperformance probabilities and one-sided significance tests.

FIGURE 20

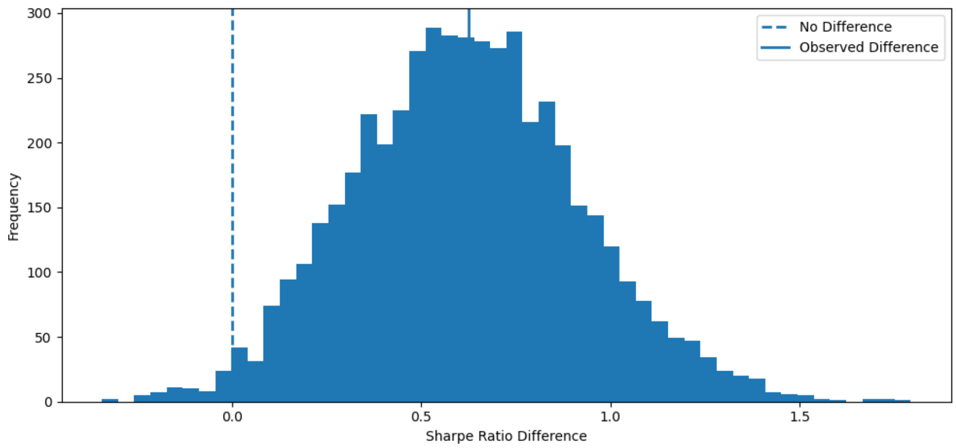

Figure 20. Bootstrap distribution of the strategy-minus-buy-and-hold Sharpe difference.

FIGURE 21

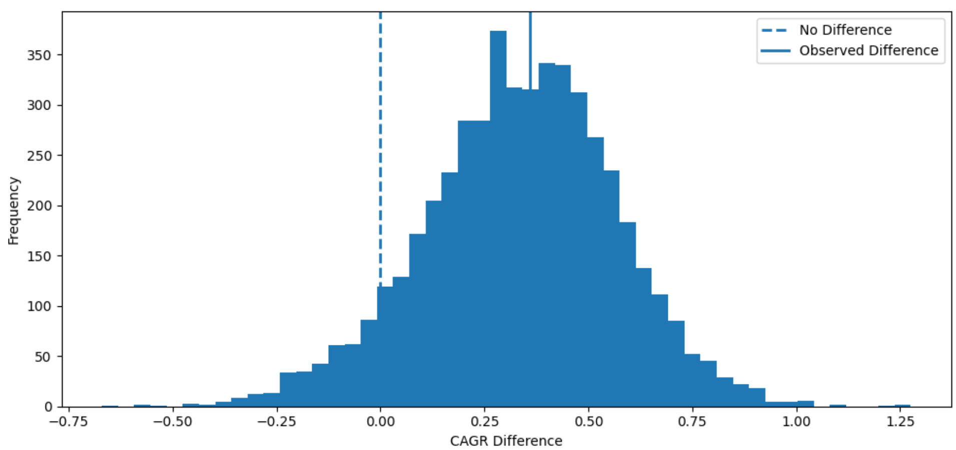

Figure 21. Bootstrap distribution of the strategy-minus-buy-and-hold CAGR difference.

Across 5,000 moving-block bootstrap samples, the strategy's Sharpe ratio exceeded buy-and-hold in 98.64% of samples, and its maximum drawdown was better in 99.28%. The corresponding one-sided p-values were 0.0138 and 0.0074. CAGR exceeded buy-and-hold in 92.18% of samples, but its p-value of 0.0784 did not meet the conventional 5% threshold. The evidence is therefore strongest for improved risk-adjusted performance and drawdown control rather than for a precisely estimated CAGR advantage.

j. Economic Interpretation

The main economic finding is that the cycle signal appears more useful for controlling exposure than for predicting every daily movement. The model spends long periods in one state, reducing unnecessary trading, and its strongest contribution comes from remaining out of the market during severe Bearish regimes. This produces lower volatility, smaller drawdowns, and stronger compounded returns.

The year-by-year results suggest that the strategy's advantage came from both participating in major Bullish periods and reducing exposure during severe declines. The most economically important example was 2022, when the Long/Cash strategy lost only 12.15% compared with a 64.27% decline for buy-and-hold. Avoiding this type of drawdown materially improved long-run compounding and allowed the strategy to recover more efficiently during subsequent market advances.

The threshold analysis further supports the finalized 3% specification. Although the 0% threshold produced higher classification accuracy, it generated weaker trading performance, demonstrating that predictive accuracy and economic value are not equivalent.

Summary of economic results

The cycle-based Long/Cash strategy generated a 62.07% CAGR compared with 26.13% for buy-and-hold, increased the Sharpe ratio from 0.69 to 1.31, reduced annualized volatility from 64.74% to 44.62%, and improved maximum drawdown from −81.53% to −51.86% — all while invested only 48.51% of the time. The advantage came from selective participation rather than continuous risk-taking.

Several limitations remain. The full cycle model uses historically identified bottoms that are not known with certainty in real time. The 30-day target creates overlapping observations, and the backtest assumes daily execution at a fixed 0.10% cost without explicitly modelling slippage, financing costs, taxes, or short-borrow constraints.

VII. Discussion and Limitations

a. Interpretation of the Main Results

The central result of this study is that Bitcoin cycle-position variables contained more useful out-of-sample information than the technical and macroeconomic feature sets tested here. The cycle model produced the strongest classification results in the original benchmark and also outperformed the finalized macroeconomic model under the 3% target. By contrast, the technical model performed poorly, and the combined model did not improve on the cycle-only specification.

This pattern suggests that Bitcoin returns may be more closely related to broad regime position than to isolated short-term indicators. Technical variables such as momentum, moving averages, and volatility can change meaning across different market phases. A momentum signal observed early in a Bullish regime may have a different implication from the same signal near the end of an extended advance. Cycle variables provide a broader context for interpreting these movements.

The trading results support the same conclusion. The model was most useful when it was used to decide whether to hold Bitcoin or remain in cash. Its value did not depend on predicting every daily move. Instead, the strategy benefited from remaining invested during favourable regimes and reducing exposure during large declines.

b. Why the Combined Model Underperformed

The combined model included technical, macroeconomic, and cycle variables, but it did not outperform the cycle-only model. Adding more information therefore did not produce a stronger forecast. One possible explanation is that many of the additional variables were correlated transformations of the same underlying price behaviour. Moving averages, momentum measures, volatility indicators, and market-return variables can all reflect overlapping information.

The larger feature set may also have increased estimation noise relative to the amount of independent data available. Although the dataset contains thousands of daily observations, those observations are drawn from only a small number of major Bitcoin regimes. A more complex model can fit relationships that appear useful within one cycle but fail to remain stable in the next. In this setting, the cycle variables may have worked better because they represented a smaller and more focused description of the market state.

This result shows that model complexity and predictive value are not the same. The strongest specification was not the one with the largest number of features, but the one that most directly measured the underlying regime being studied.

c. Neutral-Class Weakness

The largest classification weakness was the Neutral class. The finalized cycle model identified Bullish and Bearish observations much more successfully than Neutral observations, while Neutral recall was only 3.9%. Most Neutral periods were classified as either Bullish or Bearish.

This result is understandable because the Neutral class is defined by a relatively narrow range of future returns. Small changes in price can move an observation across the 3% boundary, even when the overall market environment is similar. Bullish and Bearish regimes are more distinct because they are associated with larger directional movements.

The model should therefore be interpreted mainly as a directional regime classifier rather than a precise three-state forecasting system. Its practical value comes from separating favourable from unfavourable periods, not from consistently identifying low-magnitude returns.

d. Economic Meaning of the Signal

The results from the probability analysis and the trading backtest point to the same economic interpretation. Higher predicted Bullish probability was associated with stronger subsequent 30-day returns, and the highest-confidence Bullish deciles produced the strongest average realized returns. This suggests that the model's probabilities contained useful ranking information in addition to the final class label.

The Long/Cash strategy also produced substantially stronger risk-adjusted performance than buy-and-hold while remaining invested less than half the time. Its advantage came from selective exposure rather than continuous participation. The 2022 results show this clearly: reducing exposure during a major decline had a large effect on long-run compounding.

For this reason, the cycle signal appears most useful as an exposure-management tool. It is better suited to answering when Bitcoin risk should be held than to predicting the exact size of future returns.

e. Look-Ahead and Real-Time Implementability

The most important limitation is the construction of the full cycle feature set. Variables such as Days Since Bottom, Days Until Top, Percent Through Cycle, Bull Progress, and Bear Progress depend on historically identified market turning points. These dates are clear after a cycle has ended, but they are not known with certainty in real time.

As a result, the full model should be interpreted primarily as a test of whether cycle position contains predictive information. It should not yet be treated as a fully deployable live-trading system. A real-time version would require an objective rule for identifying cycle bottoms and transitions using only information available at the time.

The halving-only model provides a partial robustness check because halving dates are observable and fixed in advance. That specification still produced positive risk-adjusted results, but it was materially weaker than the full model. This suggests that halving timing contains useful information, while the stronger performance of the full model also depends on richer market-cycle landmarks.

f. Small Number of Independent Cycles

Bitcoin has experienced only a limited number of complete historical cycles. The daily dataset contains many observations, but those observations do not represent thousands of independent market regimes. Large groups of rows belong to the same Bullish or Bearish phase and are influenced by the same underlying cycle.

This creates a risk that the model has learned patterns that are specific to a small number of past cycles. Future Bitcoin markets may differ because of institutional adoption, exchange-traded products, regulation, derivatives activity, liquidity changes, or shifts in monetary conditions. If future cycles become weaker, longer, shorter, or less connected to halvings, the relationships found here may not remain stable.

The effective sample size is therefore smaller than the raw number of daily observations suggests. The results provide evidence across several historical regimes, but they do not establish that the same pattern will persist indefinitely.

g. Overlapping Targets and Statistical Dependence

The target variable is the forward 30-day return calculated each day. This creates substantial overlap between adjacent observations. Forecasts made one day apart share 29 of their 30 future return days, so the resulting labels and realized returns are highly dependent.

This dependence means that standard statistical tests based on independent observations may overstate precision. The number of daily rows is not equal to the number of independent forecasting events. The moving-block bootstrap used in the backtest helps address this issue by resampling groups of consecutive observations rather than individual days, preserving part of the serial dependence in the data.

Even with that adjustment, overlapping targets remain an important feature of the research design. Future work could compare the current daily forecasting setup with non-overlapping monthly forecasts or alternative holding periods.

h. Data and Backtest Assumptions

The backtest uses a fixed transaction cost of 0.10% per unit of turnover and a one-day execution lag. These assumptions are more realistic than a costless same-day strategy, but they do not capture every implementation friction. The analysis does not explicitly model slippage, exchange spreads, taxes, custody costs, financing costs, or the return earned on cash.

The Long/Short strategy is especially sensitive to implementation assumptions because short Bitcoin exposure may require futures, perpetual swaps, margin, or borrowing. These instruments introduce funding payments, liquidation risk, and additional trading costs. The Long/Cash strategy is therefore the more directly implementable version of the model.

Macroeconomic data may also be affected by publication timing and later revisions. A fully real-time study would need to align each macroeconomic observation with its original release date and use the vintage that was available to investors at the time.

i. Model and Feature Limitations

The XGBoost models were estimated using a fixed parameter specification rather than an extensive nested hyperparameter search. This reduced the risk of repeatedly optimizing the model on the same test periods, but it also means that some specifications may not have been tuned to their best possible performance.

Feature importance should not be interpreted as causal evidence. A variable can rank highly because it is useful for splitting the sample, but this does not prove that it causes future Bitcoin returns. Correlated features can also divide importance among themselves, making individual rankings unstable.

The 3% target threshold was selected after comparing several alternatives. Although the threshold analysis was reported transparently, this creates some specification-selection risk. The 3% result should therefore be viewed as the best-performing tested specification rather than a universally optimal threshold.

Finally, model probabilities may not be perfectly calibrated. A predicted Bullish probability of 70% does not necessarily mean that Bullish outcomes occur exactly 70% of the time. Calibration methods could improve the interpretation of confidence levels and the design of probability-based position sizing.

j. Future Research

The most important next step is to construct cycle variables that are available in real time. This could include algorithmic bottom-identification rules, drawdown-based regime definitions, moving-average state transitions, or probabilistic estimates of cycle progress. Such variables would preserve the central cycle idea without relying on turning points identified after the fact.

Future work could also test binary targets, alternative return horizons, non-overlapping forecasts, adaptive cycle lengths, and calibrated probabilities. Rather than using a fixed Long/Cash position, position size could be linked to predicted Bullish probability or expected return. More detailed simulations could incorporate slippage, cash yields, futures funding, taxes, and exchange-specific trading costs.

The framework should also be tested on additional assets and future data. Ethereum and other large crypto assets could show whether the signal is specific to Bitcoin or reflects a broader digital-asset cycle. Most importantly, the model should be evaluated through genuine forward paper trading without revising historical turning points or thresholds after observing new outcomes.

k. Discussion Summary

The findings support the view that Bitcoin cycle position contains meaningful predictive and economic information. Cycle variables outperformed the technical and macroeconomic benchmarks, and their strongest practical value came from controlling market exposure and avoiding major drawdowns.

At the same time, the strongest specification depends partly on historical turning points that are not known with certainty in real time, and the evidence is based on a limited number of major cycles. The model should therefore be viewed as evidence for the usefulness of cycle-aware forecasting rather than as a finished live-trading system. A deployable version would require real-time cycle identification, additional independent market regimes, and prospective validation.

VIII. Conclusion

Cycle position carries information about Bitcoin's forward return distribution that standard technical and macroeconomic indicators do not fully capture — and that information is most valuable when used to decide how much risk to hold, not to forecast price.

This study tested whether Bitcoin's position in its broader market cycle helps predict future returns. Four XGBoost models were compared using technical, macroeconomic, cycle, and combined feature sets. The cycle model performed best overall. Technical indicators added little predictive value, the macro model performed better but still lagged the cycle model, and the combined model did not improve results.

Using the final ±3% target, the cycle model reached 51.26% out-of-sample accuracy and a Macro F1 score of 0.401. It was much better at identifying Bullish and Bearish periods than Neutral ones. The most useful features were measures of where Bitcoin was within the cycle, especially Percent Through Cycle, Days Since Halving, Days Until Top, and Days Since Bottom.

The trading results were stronger than the classification results alone might suggest. The Long/Cash strategy produced a 62.07% CAGR, compared with 26.13% for buy-and-hold. Its Sharpe ratio rose from 0.69 to 1.31, while volatility and maximum drawdown were both lower. The strategy was invested only 48.51% of the time, so the result mainly came from choosing when to hold Bitcoin rather than staying exposed throughout the entire sample.

The biggest benefit came from avoiding major declines. In 2022, for example, the Long/Cash strategy lost 12.15%, while buy-and-hold fell 64.27%. Avoiding losses of that size had a large effect on long-run compounding.

The model still has important limitations. Several cycle features rely on historical market bottoms and tops that are easy to identify only after the fact. The dataset also covers only a small number of major Bitcoin cycles, and the 30-day forward-return targets overlap heavily. Because of this, the results should be treated as evidence that cycle position matters, not as proof that the current model is ready for live trading.

Overall, the findings suggest that Bitcoin cycle position contains useful information that is not fully captured by standard technical or macroeconomic indicators. The next step is to build cycle features that can be measured in real time and test the strategy on future data without changing the model after new results are observed.

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Appendix

The appendix contains supplementary definitions, diagnostics, robustness results, and reproducibility details. Results already reported in the main text are not repeated unless needed to interpret a supplementary table.

A. Feature Dictionary

A.1 Technical Features

Feature	Definition	Role
EMA20 / EMA50	20- and 50-day exponential moving averages	Trend
MACD / Signal / Hist.	12–26 EMA spread, 9-day signal, and difference	Momentum
RSI	14-day relative strength index	Overbought / oversold
Stochastic oscillator	Close relative to recent high-low range	Momentum position
Bollinger bands / width	Rolling mean ± volatility bands	Price envelope
ATR	Average true range	Trading-range volatility
OBV / volume change	On-balance volume and daily volume growth	Price-volume pressure
Returns / volatility	Daily return and 7- and 30-day realized volatility	Return distribution
Range / candle vars.	Range, range %, body, body %, upper wick, lower wick	Daily price action

A.2 Macroeconomic and Cross-Asset Features

Feature	Series	Economic interpretation
Fed Funds	Federal funds effective rate	Monetary-policy stance
CPI	U.S. Consumer Price Index	Inflation
M2	U.S. M2 money stock	Liquidity
10-Year Treasury	U.S. 10-year yield	Long-term discount rate
S&P 500	Broad U.S. equity index	Risk appetite
Nasdaq	Technology-heavy equity index	Growth-asset conditions
DXY	U.S. Dollar Index	Dollar strength

A.3 Cycle Features

Feature	Definition	Caution
Days Since Bottom	Days elapsed since assigned historical bottom	Known cleanly only in hindsight

Feature	Definition	Caution
Days Until Top	Days remaining to assumed 1,064-day bull endpoint	Retrospective construction
Bull Phase	Indicator for assumed bull phase	Depends on historical cycle rule
Percent Through Cycle	Elapsed days / 1,428-day full-cycle length	Normalized cycle position
Days Since Halving	Days since most recent halving	Observable in real time
Cycle Number	Identifier for historical cycle episode	Observable once mapping is fixed
Bull Progress	Normalized progress through 1,064-day bull phase	Retrospective
Bear Progress	Normalized progress through 364-day bear phase	Retrospective

Historical bottoms: January 14, 2015; December 15, 2018; November 21, 2022. Halvings: July 9, 2016; May 11, 2020; April 20, 2024.

B. Model and Target Specification

Parameter	Final setting
Estimator	XGBoost multiclass classifier
Objective	Three-class probability classification
Trees / depth	300 trees; maximum depth 4
Learning rate	0.05
Row / feature subsampling	0.80 / 0.80
Random seed	42
Validation	Expanding-window walk-forward testing
Forecast horizon	30 calendar days
Decision rule	Class with highest predicted probability

Target Classes

Class	30-day forward-return rule	Observations	Share
Bearish	Below −3%	1,265	37.2%
Neutral	−3% to +3%	543	16.0%
Bullish	Above +3%	1,595	46.9%

The final labeled sample contains 3,403 observations from February 19, 2017 through June 14, 2026.

Feature Sets

Model	Information set
Technical	23 price- and volume-derived variables
Macroeconomic	7 macroeconomic and cross-asset variables
Cycle	8 cycle-position variables
Combined	All 38 features

C. Supplementary Predictive Results

C.1 Broader Finalized Cycle Classification Report

Class	Precision	Recall	F1	Support
Bearish	0.480	0.606	0.536	1,111
Neutral	0.184	0.039	0.064	488
Bullish	0.566	0.606	0.585	1,323
Macro average	**0.410**	**0.417**	**0.395**	**2,922**

C.2 Confusion Matrix Counts

Actual \ Predicted	Bearish	Neutral	Bullish
Bearish	673	47	391
Neutral	244	19	225
Bullish	484	37	802

This 2,922-observation report is a broader finalized output and is not the 1,826-observation common sample used in the exact cycle-versus-macro test.

D. Feature Importance and Ablation

D.1 Final Cycle Feature Importance

Feature	Average importance
Percent Through Cycle	0.1823
Days Since Halving	0.1721
Days Until Top	0.1631
Days Since Bottom	0.1333

Feature	Average importance
Bull Progress	0.1194
Bear Progress	0.1147
Cycle Number	0.0968
Bull Phase	0.0183

D.2 Cycle-Feature Ablation

Removed feature	Accuracy change	Sharpe change	CAGR change
Days Since Halving	−4.82 pp	−0.17	−15.79 pp
Cycle Number	−2.19 pp	−0.14	−12.78 pp
Percent Through Cycle	−0.82 pp	−0.09	−7.17 pp
Bull Progress	−0.27 pp	−0.09	−8.18 pp
Bear Progress	−0.44 pp	−0.15	−13.04 pp
Days Since Bottom	+1.31 pp	−0.05	−5.64 pp
Days Until Top	+0.38 pp	−0.18	−12.18 pp

A feature can reduce classification accuracy while still improving economic performance. This is why accuracy, Sharpe, and CAGR are reported together.

E. Probability and Robustness Diagnostics

E.1 Bullish-Probability Ranking

Decile	Avg. bull prob.	Avg. 30D return	Median 30D return	N	Bullish freq.
1	0.0103	−7.01%	−7.27%	192	22.40%
2	0.0380	−5.08%	−8.13%	177	19.77%
3	0.0791	1.40%	−0.66%	179	37.99%
4	0.1947	1.81%	0.86%	186	41.40%
5	0.4914	2.14%	−0.15%	179	36.87%
6	0.7250	9.39%	7.70%	183	60.11%
7	0.8970	0.56%	−0.70%	189	42.33%
8	0.9352	2.41%	1.46%	178	41.57%
9	**0.9718**	**12.18%**	**9.38%**	**198**	**66.67%**

Decile	Avg. bull prob.	Avg. 30D return	Median 30D return	N	Bullish freq.
10	0.9823	12.72%	12.69%	165	67.27%

Correlation between Bullish probability and the next 30-day return: 0.275. OLS coefficient: 0.119; $p < 0.001$; R-squared = 0.076.

E.2 Halving-Only and Threshold Checks

Test	Result	Interpretation
Halving-only feature set	CAGR 28.36%; Sharpe 0.82	Positive, materially weaker than full cycle
0% target threshold	Highest accuracy (55.73%)	Weak economic performance
3% target threshold	**Highest CAGR / Sharpe**	**Finalized specification**
5% target threshold	Competitive	Slightly weaker than 3%
10% target threshold	Lower CAGR and Sharpe	Threshold too restrictive

F. Reproducibility and Code Availability

All data preparation, feature engineering, model estimation, walk-forward testing, statistical analysis, and backtesting were completed in Python. Bitcoin and U.S. Dollar Index price series are retrieved with yfinance from Yahoo Finance; macroeconomic and equity-index series are retrieved with pandas-datareader from FRED. Technical indicators are computed with the ta library. The modelling workflow uses pandas and NumPy for data handling, XGBoost for classification, scikit-learn for evaluation, SHAP for interpretation, and matplotlib for visualization.

The complete notebook contains the data-loading steps, feature definitions, target construction, walk-forward loops, classification reports, feature-importance calculations, ablation tests, probability analysis, strategy construction, transaction-cost adjustment, and moving-block bootstrap.

Code availability statement

All code used to produce the results in this study is available from the author upon request. This statement should be replaced by a permanent repository citation if the notebook is later deposited on GitHub, Zenodo, or another archive.

Reproduction Checklist

Step	Procedure
1	Load and align Bitcoin, macroeconomic, and cross-asset series by date.
2	Construct technical, macroeconomic, and cycle features using only available information.

Step	Procedure
3	Calculate the next 30-day return and assign Bearish, Neutral, or Bullish labels.
4	Run expanding-window walk-forward estimation and store out-of-sample probabilities.
5	Choose the class with the highest probability and calculate classification metrics.
6	Shift trading signals by one day and subtract 0.10% per turnover unit.
7	Calculate CAGR, volatility, Sharpe, drawdown, exposure, and turnover.
8	Run ablation, threshold, halving-only, probability-decile, and bootstrap checks.